PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Medication adherence among allogeneic haematopoietic stem cell transplant recipients: a systematic review protocol
AUTHORS	Visintini, Chiara; mansutti, irene; palese, alvisa

VERSION 1 – REVIEW

REVIEWER	Hoegy, Delphine Hospices Civils de Lyon, pharmacy
REVIEW RETURNED	02-Aug-2022

GENERAL COMMENTS	Reviewing BMJ Open-2022-065676 Very interesting subject - thanks to propose it. It is important to publish data about medication adherence among allogeneic hematopoietic stem cell transplantation after discharge which conditioned the patient's pronostic. Has the study already started ? « last update of the literature search is 30th April 2022 » Abstract : Explications of the research & analysis strategy are missing About findings, it is important to explain how your systematic review can interest patients and caregivers, the benefits for them. Introduction : -Introduction is too long (4 pages) because presenting part of results of the review. It is not expect in an introduction section of protocol. Please remove it and wait the writing of your results and / or discussion section of the review to adress it. Instances : lines 20 to 55 page 5 ; lines 5 to 18 page 5 ; pages 6 and 7. All those elements should be approch in the introduction but not given all the results as it is done here. -Second paragraph : Why do you not talk about the 5 dimensions (described in WHO Sabaté, 2003) which influence medication adherence (condition, therapy, patient, socioeconomic factors and healthcare system) ? Figure of it would be appreciated here. Medication adherence and also non-adherence is used : The primary objective of the review would it be to prevalence rate of medication adherence or non-adherence ? -Paragraph 3 of the page 5 : If you talk about the methods to quantify ME or NA, please be the most possible exhaustive about methods. A systematic review (even if all pediatric transplantation including HSCT) are available about heterogeneous methods to quantify : this review needs to be present in your introduction, because adressing research questions of your protocol. « D Hoegy, N Bleyzac, P Robinson, Y Bertrand, C Dussart, A Janoly-Dumenil. Medication adherence in pediatric transplantation and assessment methods: a systematic review. Patient Prefer Adherence. 2019; 13:

	705-719. » Methodology : -What a pity to exclude qualitative studies, I do not understand this choice when you try to describe study available about « predictors and risk factors of medication NA » About factors influencing medication adherence (or non-adherence), it is essential from my point of view to look about facilitators as well as barriers. I recommend you to read « D Hoegy, N Bleyzac, C Rochet, H De Freminville, C Rénard, K Kébaili, Y Bertrand, C Dussart, A Janoly-Dumenil. Medication adherence after pediatric allogeneic stem cell transplantation: barriers and facilitators. Eur J Oncol Nurs. 2019; 38: 1–7. » -I recommend you to include pediatrics. There are not so much references about this population and your systematic review could discuss about particularity of pediatric medication adherence after HSCT. - Explications of the analysis strategy are missing : do you look at the type of medication ? type of immunosuppressant ? delay from transplantation ? How will you do the analysis ? - About interventions, thank you to propose to review this essential aspect. Interventions from other patients, peer patients and patients formed in education patient should not be omitted.

REVIEWER	Pai, Ahna Cincinnati Children's Hospital Medical Center, Department of Pediatrics
REVIEW RETURNED	13-Sep-2022

GENERAL COMMENTS	The authors describe a protocol to conduct a systematic review of adherence in the adult hematopoietic stem cell transplant population. This is an important and quickly evolving topic. deserving of systematic review and has the potential to inform the future clinical and scientific endeavors. Please note that the page numbers refer to the PDF pages not the pages that the authors indicated on their versions of the manuscript. The following suggestions are provided to maximize the value of the protocol paper and ultimately the review. Page 4 Line 32: The following sentence seems to be missing a word: “A high adherence to this highly complex regimen is required to avoid infections and to minimize the risk of other complications, such as Graft-versus-Host Disease (GvHD) or disease relapse.” Perhaps they mean either “High adherence….” Or “A high level of adherence…”. Page 4, Second paragraph of the introduction: The focus of this paragraph on terminology for adherence terminology is distracting. I recommend to the authors that they simply indicate and cite which definition of adherence they're going to use for this particular review. Page 5, Second paragraph. Please consider rephrasing the following sentence as it sounds like the remainder of the paper is going to report on this study. “The following cross-sectional multicentric survey enrolling 203 French adult allogeneic transplanted patients reported a NA rate of 75%, with an age of less than 50 years being a significant risk factor (p = 0.05)’. Page 7. There is another sentence that seems to not have translated well. Please rephrase as it is unclear.

	Page 9 Under, “Outcomes” and referring to secondary outcomes the authors stat: “Secondary outcomes are: (a) risk factors, barriers and predictors, and possibly related screening tools of NA.” Do they really mean screening tools or any measure of NA. Page 9. Eligibility criteria: Authors list as an eligible comparison, “Comparison: usual care (eg, consultations, counselling).’ Can other comparison groups be included and/no comparison group. Page 9. The information included in the eligibility criteria section doesn’t seem fully consistent with the paragraph on inclusion/exclusion criteria. Please reconcile. Page 11. Data collection process section. • It is recommended that the authors report on the control groups being used in the study. • It is also recommended that information on race, ethnicity and socioeconomic status be collected. Supplementary document: In the PRISMA table Item 12 provides the answer of Yes and No. Please clarify.

VERSION 1 – AUTHOR RESPONSE

bmjopen-2022-065676	
Reviewer 1	**Responses**
Very interesting subject - thanks to propose it. It is important to publish data about medication adherence among allogeneic hematopoietic stem cell transplantation after discharge which conditioned the patient’s pronostic. Has the study already started ? « last update of the literature search is 30th April 2022 » Abstract : Explications of the research & analysis strategy are missing About findings, it is important to explain how your systematic review can interest patients and caregivers, the benefits for them. Introduction :	Dr. Delphine Hoegy, thank you very much for your interest in our topic and review and for your detailed considerations and comments, that we really appreciated. We hope to provide you satisfactory answers. The literature search has already started, and the review is now in progress; literature search has been updated to 10th May 2022 (we revised this data in the manuscript). Abstract: Thank you, we added this information in the ‘Methods and analysis’ and ‘Ethics and dissemination’ sections.

-Introduction is too long (4 pages) because presenting part of results of the review. It is not expect in an introduction section of protocol. Please remove it and wait the writing of your results and / or discussion section of the review to adress it. Instances : lines 20 to 55 page 5 ; lines 5 to 18 page 5 ; pages 6 and 7. All those elements should be approch in the introduction but not given all the results as it is done here. -Second paragraph : Why do you not talk about the 5 dimensions (described in WHO Sabaté, 2003) which influence medication adherence (condition, therapy, patient, socioeconomic factors and healthcare system) ? Figure of it would be appreciated here. Medication adherence and also non-adherence is used : The primary objective of the review would it be to prevalence rate of medication adherence or non-adherence ? -Paragraph 3 of the page 5 : If you talk about the methods to quantify ME or NA, please be the most possible exhaustive about methods. A systematic review (even if all pediatric transplantation including HSCT) are available about heterogeneous methods to quantify : this review needs to be present in your introduction, because adressing research questions of your protocol. « D Hoegy, N Bleyzac, P Robinson, Y Bertrand, C Dussart, A Janoly-Dumenil. Medication adherence in pediatric transplantation and assessment methods: a systematic review. Patient Prefer Adherence. 2019; 13: 705-719. » Methodology : -What a pity to exclude qualitative studies, I do not understand this choice when you try to describe study available about « predictors and risk factors of medication NA » About factors influencing medication adherence (or non-adherence), it is essential from my point of view to look about facilitators as well as barriers. I recommend you to read « D Hoegy, N Bleyzac, C Rochet, H De Freminville, C Rénard, K Kébaili, Y Bertrand, C Dussart, A Janoly-Dumenil. Medication adherence after pediatric allogeneic stem cell transplantation: barriers and facilitators. Eur J Oncol Nurs. 2019; 38: 1–7. » -I recommend you to include pediatrics. There are not so much references about this population and your systematic review could discuss about particularity of pediatric medication adherence after	Introduction: The introduction has been extensively revised according to the suggestion. Specific data have been removed in order to establish an overview of the knowledge available as produced to date by primary and secondary studies in the field. Specifically, with regards the feedback received: - we considered your suggestions and removed some parts of the introduction: our intention was to give as much data as possible to the reader and not anticipate the review's findings. From the initial XXX words in the current form the introduction includes XXX words - thank you for this suggestion. We added the five dimensions from WHO (2003), anticipating the theme of non-adherence in the transplanted population. The primary aim of the review is the prevalence rate of medication adherence (we will convert non-adherence % rates - where reported - into adherence % rates) -thank you very much to have given us the opportunity to revise our protocol paying attention to the methodological aspects of adherence measurement. Thus, we cited this review about paediatrics and reported the main findings in our introduction. Methodology: - we will include only quantitative data as stated in the review registration in PROSPERO. However, we tried to research if there were any qualitative studies until 10th of May 2022

HSCT. - Explications of the analysis strategy are missing : do you look at the type of medication ? type of immunosuppressant ? delay from transplantation ? How will you do the analysis ? - About interventions, thank you to propose to review this essential aspect. Interventions from other patients, peer patients and patients formed in education patient should not be omitted.	regarding adult population but no study emerged. Respect to facilitators and barriers, we agreed with your proposal and will considered both these sides in the review; so, we inserted your citation and the one by Posluszny *et al.* (2022) in the related paragraph of the introduction. - we decided to exclude paediatrics because of differences in psychological answers to the disease, medication posology (and eg in flavor or form) and other aspects between paediatrics and adults, as emerged from your interesting review (Hoegy *et al.,* 2019) that we read carefully. Moreover, whereas in adults only the patient is considered as the subject of process of care, in the paediatric setting the days child-parent must be considered, since it is more than likely that parents have a significant role in the medication administration and other self-management processes both inpatient and outpatient. In addition, facilitators and barriers at individual, family, community, and health care system level could be different (*Morrison et al., 2017*). - yes, we would like to report adherence rates in general and then showing rates for immunosuppressors and non-immunosuppressant therapy. Considering the heterogeneity in medication adherence measurements and interventions, only a qualitative synthesis will be provided. However, we will show the minimum and maximum prevalence rate among the studies, and find the median and medium data. We add this information in the 'data synthesis' section - thank you very much for this suggestion. We hope to find interventions as these proposed among the included studies, meantime we made a suggestion about that in the introductory part.
Reviewer 2	**Responses**
The authors describe a protocol to conduct a systematic review of adherence in the adult hematopoietic stem cell transplant population. This	Dear Prof. Ahna Pai,

is an important and quickly evolving topic. deserving of systematic review and has the potential to inform the future clinical and scientific endeavors. Please note that the page numbers refer to the PDF pages not the pages that the authors indicated on their versions of the manuscript. The following suggestions are provided to maximize the value of the protocol paper and ultimately the review.	Thank you very much for your consideration and feedback, and for having given us the possibility to revise the manuscript answering to the issues requested. We really appreciated it.
Page 4 Line 32: The following sentence seems to be missing a word: “A high adherence to this highly complex regimen is required to avoid infections and to minimize the risk of other complications, such as Graft-versus-Host Disease (GvHD) or disease relapse.” Perhaps they mean either “High adherence….” Or “A high level of adherence…”.	
Page 4, Second paragraph of the introduction: The focus of this paragraph on terminology for adherence terminology is distracting. I recommend to the authors that they simply indicate and cite which definition of adherence they're going to use for this particular review.	Thank you, it was an oversight. We added ‘high level’ to the phrase. Final phrase: ‘A high level of adherence to this highly complex regimen is required […]’.
Page 5, Second paragraph. Please consider rephrasing the following sentence as it sounds like the remainder of the paper is going to report on this study. “The following cross-sectional multicentric survey enrolling 203 French adult allogeneic transplanted patients reported a NA rate of 75%, with an age of less than 50 years being a significant risk factor (p = 0.05)’.	We are afraid that this paragraph could be distracting; however, we are going to maintain these lines regarding the definition of adherence and, secondly, our adapted. We considered important to give the complete definition of the construct.
Page 7. There is another sentence that seems to not have translated well. Please rephrase as it is unclear.	Thank you for this suggestion. We rephrased and synthesized the sentence, hoping now it is better understood. Final phrase: ‘A higher rate of MNA emerged from the cross-sectional multicentric survey enrolling 203 French adult allogeneic hematopoietic stem cell transplant recipients, where the rate was of 75%’.
Page 9 Under, “Outcomes” and referring to secondary outcomes the authors stat: “Secondary outcomes are: (a) risk factors, barriers and predictors, and possibly related screening tools of NA.” Do they really mean screening tools or any measure of NA.	
Page 9. Eligibility criteria: Authors list as an eligible comparison, “Comparison: usual care (eg, consultations, counselling).’ Can other comparison groups be included and/no comparison group.	We have submitted the manuscript to a proof-reading copy-editing English service. We hope that the manuscript is now clearer of meaning and well understood.
Page 9. The information included in the eligibility criteria section doesn’t seem fully consistent with	We intended screening as the measurement is

the paragraph on inclusion/exclusion criteria. Please reconcile. Page 11. Data collection process section. • It is recommended that the authors report on the control groups being used in the study. • It is also recommended that information on race, ethnicity and socioeconomic status be collected. Supplementary document: In the PRISMA table Item 12 provides the answer of Yes and No. Please clarify.	conducted before HSCT (eg SIPAT score), as predictive of a possible future MNA post HSCT. We found examples from the solid organ transplants, in order to conduct interventions (eg psychological, educational) before HSCT to reduce the risk of MNA Thank you for this suggestion. We added 'peer-education, and/or no comparison group' to the sentence. Sorry if this information seemed not matched. We revised the two paragraphs. Thank you. We considered your proposal and added this information in the 'data collection process' section. Sorry, it was a copying error from the original file. The right answer for Item 12 is No. Thank you again for your valuable feedback

VERSION 2 – REVIEW

REVIEWER	Hoegy, Delphine Hospices Civils de Lyon, pharmacy
REVIEW RETURNED	21-Oct-2022

GENERAL COMMENTS	Introduction : About findings, it is important to explain how your systematic review

	can interest patients and caregivers, the benefits for them. Methodology : -Thank you for your explanations about not including pediatrics : I agree. Please add this information in limitations section and their explanation

REVIEWER	Pai, Ahna Cincinnati Children's Hospital Medical Center, Department of Pediatrics
REVIEW RETURNED	19-Oct-2022

GENERAL COMMENTS	In general, the authors were responsive to the reviewers recommendations. I have a couple of additional comments to further bolster this protocol paper. I recommend that the authors choose a term to consistently use throughout the manuscript to refer to hematopoietic stem cell transplant. At various points throughout the paper they use bone marrow transplant to refer to hematopoietic stem cell transplant. I recommended in the first review that the authors remove the reference to the term compliance. I understand that they would like to retain this text. Since they are keeping the text, I recommend that they revise the sentence to make it explicit that they are comparing the term compliance to the term adherence and that adherence is the preferred term. I appreciate you adding the following sentence in response to reviewer 1, ”According to the available evidence, the use of self-report approaches combined with objective methods has been recommended to prevent heterogeneity in prevalence rates.”. However, rephrasing the paragraph to clarify how you are preventing heterogeneity in prevalence rates would be helpful. Do you mean heterogeneity in prevalence rates across studies?

VERSION 2 – AUTHOR RESPONSE

Reviewer 1	**Responses**
Introduction : About findings, it is important to explain how your systematic review can interest patients and caregivers, the benefits for them. Methodology : -Thank you for your explanations about not including pediatrics : I agree. Please add this information in limitations section and their	Dr. Delphine Hoegy, thank you very much for your interest in and appreciation of our major revision. We added the information requested at the end of the ‘introduction’ section before the presentation of the aims. Thank you. We added the choice of exclusion of the paediatric population from our review at the beginning of the ‘DISCUSSION’ section.

explanation	
Reviewer 2	**Responses**
In general, the authors were responsive to the reviewers recommendations. I have a couple of additional comments to further bolster this protocol paper.	Dear Prof. Ahna Pai, thank you very much for your consideration and valuable comments.
I recommend that the authors choose a term to consistently use throughout the manuscript to refer to hematopoietic stem cell transplant. At various points throughout the paper they use bone marrow transplant to refer to hematopoietic stem cell transplant.	Sorry, we wanted to lighten the manuscript from the term 'HSCT' using 'BMT', although we were aware of the difference. Thus, we have used 'HSCT' throughout the manuscript; thanks for this clarification.
I recommended in the first review that the authors remove the reference to the term compliance. I understand that they would like to retain this text. Since they are keeping the text, I recommend that they revise the sentence to make it explicit that they are comparing the term compliance to the term adherence and that adherence is the preferred term.	Yes, we retained the reference to the term compliance; thus, we revised the related sentence in the introduction, and we hope that now it is more comprehensible. Thank you for this suggestion.
I appreciate you adding the following sentence in response to reviewer 1, "According to the available evidence, the use of self-report approaches combined with objective methods has been recommended to prevent heterogeneity in prevalence rates.". However, rephrasing the paragraph to clarify how you are preventing heterogeneity in prevalence rates would be helpful. Do you mean heterogeneity in prevalence rates across studies?	Thank you. Yes, we meaned that, thus we added 'across the studies'.

VERSION 3 – REVIEW

REVIEWER	Hoegy, Delphine Hospices Civils de Lyon, pharmacy
REVIEW RETURNED	08-Nov-2022

GENERAL COMMENTS	No further comments.

REVIEWER	Pai, Ahna Cincinnati Children's Hospital Medical Center, Department of Pediatrics

REVIEW RETURNED	14-Nov-2022

GENERAL COMMENTS	The authors have adequately address previous comments on the review.